Supplementary Information: Power Balance & CAD Geometry

Manuscript ID: Fusion-Arch-001 (Paper VII)

ESI Section 1: Detailed Reactor System Power Balance Derivation

The reactor fusion power balance follows 0D volume-integrated DT reaction rates. For P_aux = 50.0 MW and Q = 11.0, total fusion power P_fusion = 550.0 MW. Alpha particle heating P_alpha = 110.0 MW and neutron power P_neutron = 440.0 MW. Gross electric power output P_elec_gross = 281.75 MW based on a Brayton thermal conversion efficiency ηth = 0.45. Accounting for recirculating plant power P_recirc = 85.0 MW, net electric output P_elec_net = 196.75 MW.

ESI Section 2: CAD Geometry & Spatial Nodal Alignment Specifications

Conformal spatial mesh nodal alignment maintains interpolation gradient errors below 0.1% across all boundaries: (1) Liquid Metal / Tungsten Armor Interface; (2) DCLL SiCf/SiC Flow Channel Inserts; (3) Cryogenic Structural Vault Keyways (Inconel 718); (4) High-Field REBCO TF Conductors.